Module 2 – Descriptive Statistics, Application-Based Problems

1. The Canmark Research Center Airport Customer Satisfaction Survey uses an online questionnaire to provide airlines and airports with customer satisfaction ratings for all aspects of the customers' flight experience (airportsurvey website, July, 2012). After completing a flight, customers receive an e-mail asking them to go to the website and rate a variety of factors, including the reservation process, the check-in process, luggage policy, cleanliness of gate area, service by flight attendants, food/beverage selection, on-time arrival, and so on. A five-point scale, with Excellent (E), Very Good (V), Good (G), Fair (F), and Poor (P), is used to record customer ratings. Assume that passengers on a Delta Airlines flight from Myrtle Beach, South Carolina, to Atlanta, Georgia, provided the following ratings for the question, "Please rate the airline based on your overall experience with this flight." The sample ratings are shown below.(Airsurvey_Dataset)
   a. Use a percent frequency distribution and a bar chart to summarize these data. What do these summaries indicate about the overall customer satisfaction with the Delta flight?
   b. The online survey questionnaire enabled respondents to explain any aspect of the flight that failed to meet expectations. Would this be helpful information to a manager looking for ways to improve the overall customer satisfaction on Delta flights? Explain.

2. The Financial Times/Harris Poll is a monthly online poll of adults from six countries in Europe and the United States. The poll conducted in January 2008 included 1015 adults. One of the questions asked was, "How would you rate the Federal Bank in handling the credit problems in the financial markets?" Possible responses were Excellent, Good, Fair, Bad, and Terrible (Harris Interactive website, January 2008). The 1015 responses for this question can be found in the data file named FedBank (Fedbank data).

   a. Construct a frequency distribution.
   b. Construct a percent frequency distribution.
   c. Construct a bar chart for the percent frequency distribution.
   d. Comment on how adults in the United States think the Federal Bank is handling the credit problems in the financial markets.

3. The Dow Jones Industrial Average (DJIA) underwent one of its infrequent reshufflings of companies when General Motors and Citigroup were replaced by Cisco Systems and Travelers (The Wall Street Journal, June 8, 2009). At the time, the prices per share for the 30 companies in the DJIA were as follows(**DJIAPrices)**:

   a. What is the highest price per share? What is the lowest price per share?
   b. Using a class width of 10, develop a frequency distribution for the data.
   c. Prepare a histogram. Interpret the histogram, including a discussion of the general shape of the histogram, the midprice range, and the most frequent price range.
   d. Use the The Wall Street Journal or another newspaper to find the current price per share for these companies. Prepare a histogram of the data and discuss any changes since June 2009. What company has had the largest increase in the price per share? What company has had the largest decrease in the price per share?

4. Fortune provides a list of America's largest corporations based on annual revenue. Shown below are the 50 largest corporations, with annual revenue expressed in billions of dollars (CNN Money website, January 15, 2010). Summarise the data by constructing the following:

a. A frequency distribution (classes 0–49, 50–99, 100–149, and so on).
b. A relative frequency distribution.
c. A cumulative frequency distribution.
d. A cumulative relative frequency distribution.
e. What do these distributions tell you about the annual revenue of the largest corporations in America?
f. Show a histogram. Comment on the shape of the distribution.
g. What is the largest corporation in America and what is its annual revenue?